Practice 4. Probability Distributions

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What is a distribution?

A probability distribution describes how probabilities are distributed or shared among the different possible values that a random variable can take.

A distribution answers the following question: ## How likely is each possible outcome of a random experiment?

What are they used for?

1. Modeling uncertain phenomena
2. Predicting possible outcomes
3. Making data-driven decisions

Types of distributions

Discrete: Binomial, Poisson, Hypergeometric (Example of variable: Number of children, number of errors) Continuous: Normal, Exponential, Uniform (Example of variable: Weight, Height, Temperature)

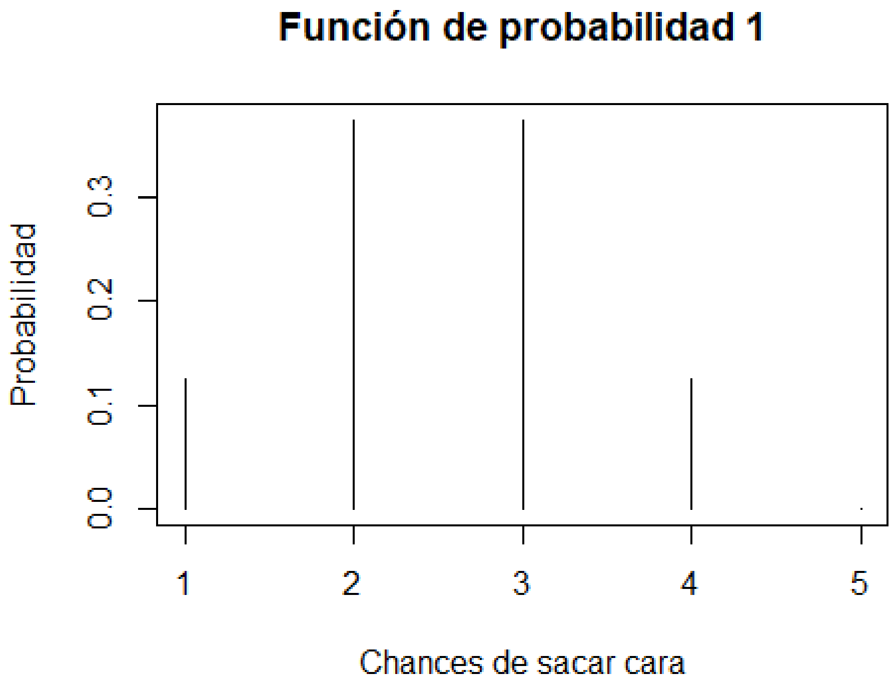
Research: Real-life examples for each type of distribution. Add a graph representing each type of distribution.

#1. Example of Binomial Distribution Description: This type of distribution is used, for example, in the experiment of tossing a coin N number of times.

```
# Example: How much do I need to get 3/5 heads?
total <- 5
good <- 3
pro <- 0.5
fail <- 0.5

probability_1 <- (factorial(total) / (factorial(good) * factorial(total-good))) * (pro**good) * (fail**(total-good))

cat("The chances to get 3/5 heads is: ", probability_1*100, "%")

## The chances to get 3/5 heads is:  31.25 %
```

Graphic.

```r
# Example of visualization
plot(dbinom(0:4, 3, 0.5), type="h",xlab="Chances de sacar cara",ylab="Probabilidad",main="Función de probabilidad 1")
```

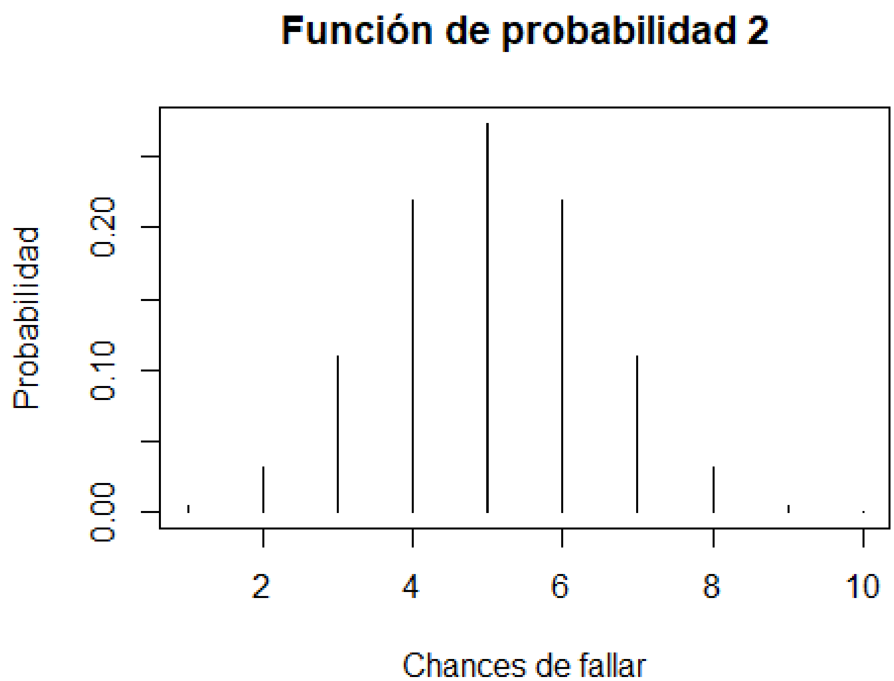


2. True or false questions

```r
# Example: get 10 times wrong t/f questions
total <- 10
good <- 10
pro <- 0.5
fail <- 0.5

probability_2 <- (factorial(total) / (factorial(good) * factorial(total-good))) * (pro**good) * (fail**(total-good))

cat("The chances to get 10 wrong questions is: ", probability_2*100, "%")

## The chances to get 10 wrong questions is:  0.09765625 %
```

Graphic.

```r
# Example of visualization
plot(dbinom(0:9, 8, 0.5), type="h",xlab="Chances de fallar",ylab="Probabilidad",main="Función de probabilidad 2")
```

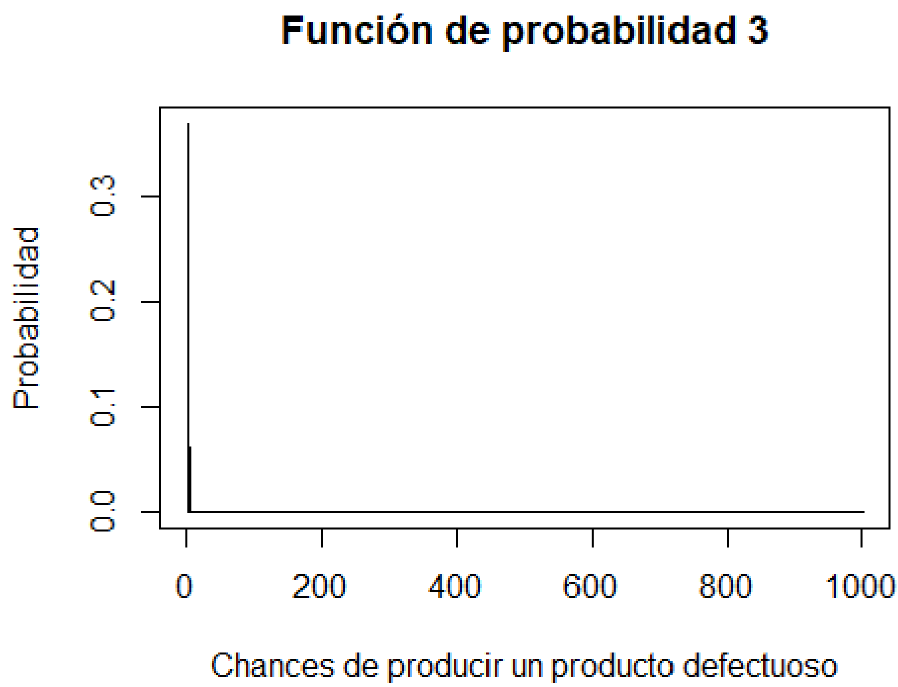


3.Production

```
# Example: Get 100 defective products over 1000 products with an 99%
efficiency
total <- 1000
good <- 100
pro <- 0.01
fail <- 0.99

probability_3 <- (factorial(total) / (factorial(good) * factorial(total-
good))) * (pro**good) * (fail**(total-good))

cat("The chances to get 100 defective products is: ", probability_3*100,
"%")

## The chances to get 100 defective products is:  NaN %
```

Graphic.

```
# Example of visualization
plot(dbinom(0:1000, 100, 0.01), type="h",xlab="Chances de producir un
producto defectuoso",ylab="Probabilidad",main="Función de probabilidad
3")
```

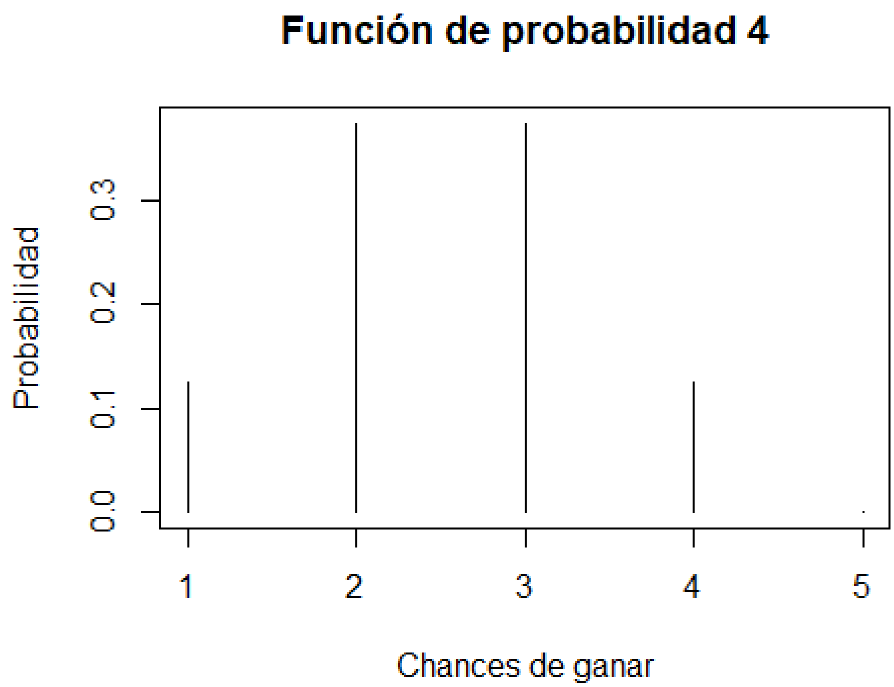


4. Soccer matches (no ties included)

```
# Example: Win 4 out of 5 matches (same prob, no diff)
total <- 5
good <- 4
pro <- 0.5
fail <- 0.5

probability_4 <- (factorial(total) / (factorial(good) * factorial(total-
good))) * (pro**good) * (fail**(total-good))

cat("The chances to win 4 matches is: ", probability_4*100, "%")

## The chances to win 4 matches is:  15.625 %
```

Graphic.

```
# Example of visualization
plot(dbinom(0:4, 3, 0.5), type="h",xlab="Chances de
ganar",ylab="Probabilidad",main="Función de probabilidad 4")
```

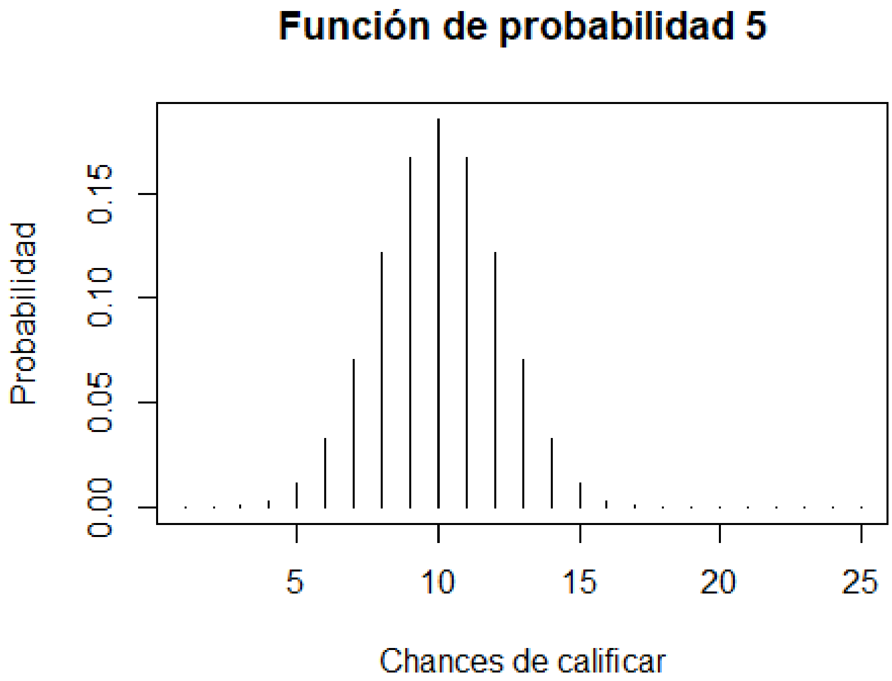


5. qualifications

```
# Example: Approve (chances are over 70%) on a 25 questions of true/false
(all blind)
total <- 25
good <- 18
pro <- 0.5
fail <- 0.5

probability_5 <- (factorial(total) / (factorial(good) * factorial(total -
good))) * (pro**good) * (fail**(total-good))

cat("The chances to qualify is: ", probability_5*100, "%")

## The chances to qualify is:  1.432598 %
```

Graphic.

```
# Example of visualization
plot(dbinom(0:24, 18, 0.5), type="h",xlab="Chances de
calificar",ylab="Probabilidad",main="Función de probabilidad 5")
```

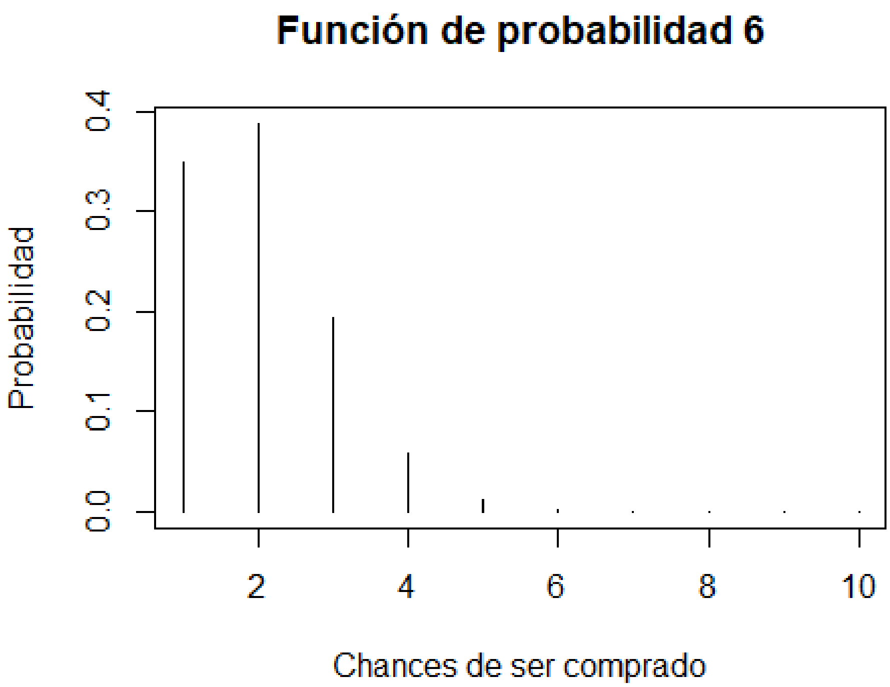


6. Sells

```
# Example Chances to buy 1 time a product after 10 clients (with a 10%
chance)
total <- 10
good <- 1
pro <- 0.1
fail <- 0.9

probability_6 <- (factorial(total) / (factorial(good) * factorial(total -
good))) * (pro**good) * (fail**(total-good))

cat("The chances to get bought is: ", probability_6*100, "%")

## The chances to get bought is:  38.74205 %
```

Graphic.

```
# Example of visualization
plot(dbinom(0:9, 10, 0.1), type="h",xlab="Chances de ser
comprado",ylab="Probabilidad",main="Función de probabilidad 6")
```


Función de probabilidad 6
Probabilidad
0.0
0.1
0.2
0.3
0.4
2
4
6
8
10
Chances de ser comprado